

Machine Learning vs. Deep Learning

A Case Study Comparison

Machine Learning Example: Email Spam Detection

In the lesson, spam detection is used as a classic example of machine learning, built on algorithms like Logistic Regression or Decision Trees. Instead of the computer figuring things out entirely on its own, a person first decides which characteristics of an email actually matter — how often certain words show up, whether the message contains links, what the sender's address looks like. Those features get fed into the model along with a set of emails that have already been labeled spam or not spam, and from there the algorithm learns to sort new messages on its own.

You can see this same idea play out in real inboxes. Gmail and Outlook, especially in their earlier years, leaned on this exact approach: classifiers trained on labeled email data using features like keyword frequency, sender reputation, and link count to separate junk mail from everything else.

Why Machine Learning Fits This Problem

Spam is a good match for ML because the things that actually give it away — certain words, suspicious senders, a flood of links — are things a person can identify without too much trouble. There's no need for a model to dig through layers of abstraction to find the pattern; it's already sitting on the surface. Because of that, algorithms like logistic regression or decision trees can get quite accurate without needing an enormous training set. They're also cheap to run, which matters when a filter has to check every single incoming email, and they're easy to interpret, so if a legitimate email gets flagged by mistake, it's possible to trace back why.

Why Deep Learning Would Be Overkill Here

Deep learning could technically be applied to spam filtering, but it wouldn't really be worth it. Training a neural network takes far more data and computing power, and for a problem this well-defined, that extra cost doesn't translate into much of an accuracy improvement. It also makes the decision-making harder to explain — neural networks don't offer the same kind of clear reasoning a decision tree does, which is a real downside if a company needs to justify why a message was blocked.

That said, it's worth noting that spam has gotten more sophisticated over the years. As phishing attempts start mimicking real writing styles and legitimate-looking correspondence, some companies have actually started leaning on deep learning to catch patterns that simple keyword rules would miss. But for the more straightforward, keyword-and-sender version of spam detection described in the lesson, traditional ML still does the job efficiently.

Deep Learning Example: Autonomous Driving

The lesson's deep learning example is autonomous driving, which relies on CNNs or deep reinforcement learning. Unlike the spam example, there's no human sitting down beforehand to define which features matter. Instead, the model is handed raw sensor data — camera footage, LIDAR readings — and has to learn on its own how to recognize pedestrians, traffic lights, road signs, and everything else it needs to drive safely in real time.

Waymo is probably the best real-world example of this working at scale. As of early 2026, the company runs fully autonomous, Level 4 ride-hailing service across ten U.S. metro areas and is completing more than 450,000 paid driverless rides every week, with plans to reach a million rides a week by the end of the year. Underneath all of that are CNNs processing camera and LIDAR data on the fly, identifying vehicles, pedestrians, and traffic signals — basically the same CNN-based process described in the lesson, just operating at massive scale.

Source: [AIMultiple](#) — “Top 50 Deep Learning Use Case & Case Studies”

Why Deep Learning Fits This Problem

Self-driving cars deal with a level of complexity that just doesn't work for hand-crafted features. There's no realistic way for a person to sit down and define rules for every possible pedestrian pose, lighting condition, or road layout — there are just too many variables. CNNs get around this by learning those features directly from the raw pixels themselves. Waymo also has the kind of massive, constantly growing dataset from real and simulated driving that deep learning actually needs to perform well, and the relationships between all those sensor inputs and split-second driving decisions are far too nonlinear and layered for a simpler model to capture.

Why Traditional Machine Learning Wouldn't Work Here

A simpler ML model just isn't built for this kind of problem. Trying to hand-code features for every lighting condition, weather pattern, or road scenario a car might run into would be practically impossible — there are too many variables to account for manually. Models like decision trees or SVMs also don't scale well to the kind of high-dimensional, continuous sensor data self-driving cars generate. And because the stakes are so high — NHTSA opened an investigation in early 2026 after a Waymo vehicle struck a child near a school — this is exactly the kind of situation where the extra precision deep learning offers isn't optional, it's necessary.

Summary

Looking at spam detection and autonomous driving side by side really shows how different these two problems are. Spam detection works well with machine learning because the important features are

things a person can point to, and the data itself is fairly structured. Autonomous driving needs deep learning because the data is raw and messy — camera images, LIDAR point clouds — and the patterns involved are far too complex for anyone to define by hand. In the end, it's less about which approach is “smarter” and more about which one actually fits the shape of the problem in front of you.